Spatial Exploratory Analyses

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#Load Packages

```
options(repos = c(CRAN = "https://cloud.r-project.org/"))
```

```
install.packages("readxl")
```

```
##
## The downloaded binary packages are in
##  /var/folders/lg/6yy176k544v684zp47118_wh0000gn/T//Rtmp5uXnli/downloaded_packages
```

```
library(readxl)
```

```
heatmap <- read_excel("~/Downloads/NIRD 20230130 Database_Hackathon.xlsx", sheet = 2)
```

```
install.packages(c("tidygeocoder", "ggplot2", "sf", "leaflet"))
```

```
##
## The downloaded binary packages are in
##  /var/folders/lg/6yy176k544v684zp47118_wh0000gn/T//Rtmp5uXnli/downloaded_packages
```

```
library(tidygeocoder)
library(ggplot2)
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(leaflet)
```

```
#Retrieve spatial data for plotting
geocoded_data <- heatmap %>%
  geocode(street = ADDRESS, city = CITY, state = STATE_FULL, postalcode = ZIP_CODE, method = "osm", lat
```

```
## Passing 635 addresses to the Nominatim single address geocoder
```

```
## Query completed in: 638.3 seconds
```

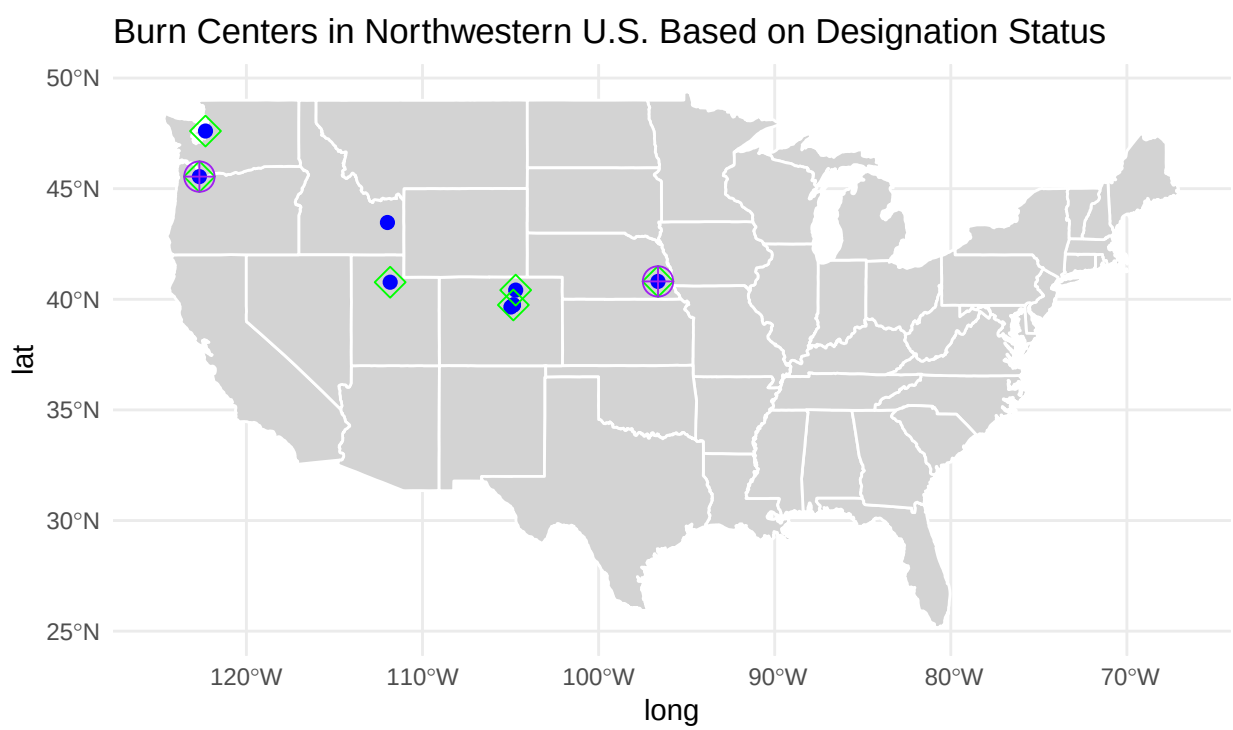
```r
###Subset Geocoded Data by State for NW United States
Montana <- geocoded_data[geocoded_data$STATE_FULL == "Montana",]

#Billings clinic is lvI trauma center post-creation of NIRD dataset -> recode it as LVI
Montana[3, 16] <- 1
Montana[3, 17] <- NA
Idaho <- geocoded_data[geocoded_data$STATE_FULL == "Idaho",]
SD <- geocoded_data[geocoded_data$STATE_FULL == "South Dakota",]
###some missing data, manually input longitutde and latitude using latlong.net geocode tools
ND <- geocoded_data[geocoded_data$STATE_FULL == "North Dakota",]
ND[3, 24] <- 46.832719
ND[3, 25] <- -96.798406
ND[4, 24] <- 46.874075
ND[4, 25] <- -96.788057
WY <- geocoded_data[geocoded_data$STATE_FULL == "Wyoming",]
###some missing data for WY, manually input longitutde and latitude
WA <- geocoded_data[geocoded_data$STATE == "WA",]
WA[3, 24] <- 47.133160
WA[3, 25] <- -122.492532
OR <- geocoded_data[geocoded_data$STATE == "OR",]
###some missing data for OR, manually input longitutde and latitude
OR[4,24] <- 44.081564
OR[4, 25] <- -123.026697
UT <- geocoded_data[geocoded_data$STATE_FULL == "Utah",]
###some missing data for UT, manually input longitutde and latitude
UT[1, 24] <- 40.686244
UT[1, 25] <- -111.857494
UT[2, 24] <- 40.632114
UT[2, 25] <- -111.899611
CO <- geocoded_data[geocoded_data$STATE_FULL == "Colorado",]
###some missing data for CO, manually input longitutde and latitude
CO[6, 24] <- 39.576425
CO[6, 25] <- -104.986152
CO[10, 24] <- 39.534390
CO[10, 25] <- -104.898072
NE <- geocoded_data[geocoded_data$STATE_FULL == "Nebraska",]
##Combine Clinics for Northwest Region
west_states <- rbind(Montana, Idaho, SD, ND, WY, WA, OR, UT, CO, NE)
#subset any clinics with burn beds, regardless of designation
burn_beds <- west_states[which(west_states$BURN_BEDS != is.na(NA)),]
#subset aba-verified burn centers
burn_centers_ABA <- west_states[which(west_states$ABA_VERIFIED == "Yes"),]
#subset state-designated burn centers
burn_centers_SD <- west_states[which(west_states$BC_STATE_DESIGNATED == "Yes"),]
#merge any type of burn care (burn beds, aba-verified centers, or state-designated burn centers)
burn_capacity <- rbind(burn_beds, burn_centers_ABA, burn_centers_SD)
#ensure only unique centers are present
burn_capacity <- unique(burn_capacity)
#subset burn centers designated for adults
burn_capacity_adults <- burn_capacity[which(burn_capacity$BURN_ADULT == 1), ]
#subset burn centers designated for peds & adults
burn_capacity_peds <- burn_capacity[which(burn_capacity$BURN_PEDS == 1), ]
#subset LVI trauma centers with pediatric and/or adult capacity
```

```r
tc_1 <- west_states[which(west_states$ADULT_TRAUMA_L1 == 1 | west_states$PEDS_TRAUMA_L1 == 1),]
#subset LVII trauma centers with pediatric and/or adult capacity
tc_2 <- west_states[which(west_states$ADULT_TRAUMA_L2 == 1 | west_states$PEDS_TRAUMA_L2 == 1 ),]
#merge LVI and LVII trauma centers with pediactric and/or adult capacity
tc_12 <- rbind(tc_1, tc_2)
#ensure only unique centers are present
tc_12 <- unique(tc_12)
tc_12_mt <- tc_12[which(tc_12$STATE_FULL == "Montana"),]
#subset LVI and LVII trauma centers based on adult trauma capacity
tc_12_adults <- tc_12[which(tc_12$TRAUMA_ADULT == 1), ]
tc_12_adults_mt <- tc_12_mt[which(tc_12_mt$TRAUMA_ADULT == 1), ]
#subset LVI and LVII trauma centers based on pediatric trauma capacity
tc_12_peds<- tc_12[which(tc_12$TRAUMA_PEDS == 1), ]
tc_12_peds_mt <- tc_12[which(tc_12_mt$TRAUMA_PEDS == 1), ]
###manually create geocoded data for county seats in four counties of interest in MT
counties_df <- data.frame(
  Name = c("Missoula", "Kalispell", "Hamilton", "Polson"),
  longitude = c(-113.9940, -114.3168, -114.1548, -114.1631),
  latitude = c(46.8721, 48.1920, 46.2471, 47.6932))
loc_counties <- st_as_sf(counties_df, coords = c("longitude", "latitude"), crs = 4326)
```

```r
#All Clinics (trauma centers and burn centers)
loc_sf_west <- st_as_sf(west_states, coords = c("longitude", "latitude"), crs = 4326)
#Burn Capacity by Designation
loc_sf_beds <- st_as_sf(burn_beds, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_aba <- st_as_sf(burn_centers_ABA, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_sd <- st_as_sf(burn_centers_SD, coords = c("longitude", "latitude"), crs = 4326)
#Total Burn Capacity by Age
loc_sf_burn <- st_as_sf(burn_capacity, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_burn_adults <- st_as_sf(burn_capacity_adults, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_burn_peds <- st_as_sf(burn_capacity_peds, coords = c("longitude", "latitude"), crs = 4326)
#Total Trauma Capacity
loc_sf_tc1<- st_as_sf(tc_1, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_tc2 <- st_as_sf(tc_2, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_tc12 <- st_as_sf(tc_12, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_tc12_mt <- st_as_sf(tc_12_mt, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_tc_adults <- st_as_sf(tc_12_adults, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_tc_adults_mt <- st_as_sf(tc_12_adults_mt, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_tc_peds <- st_as_sf(tc_12_peds, coords = c("longitude", "latitude"), crs = 4326)
loc_sf_tc_peds_mt <- st_as_sf(tc_12_peds_mt, coords = c("longitude", "latitude"), crs = 4326)
```

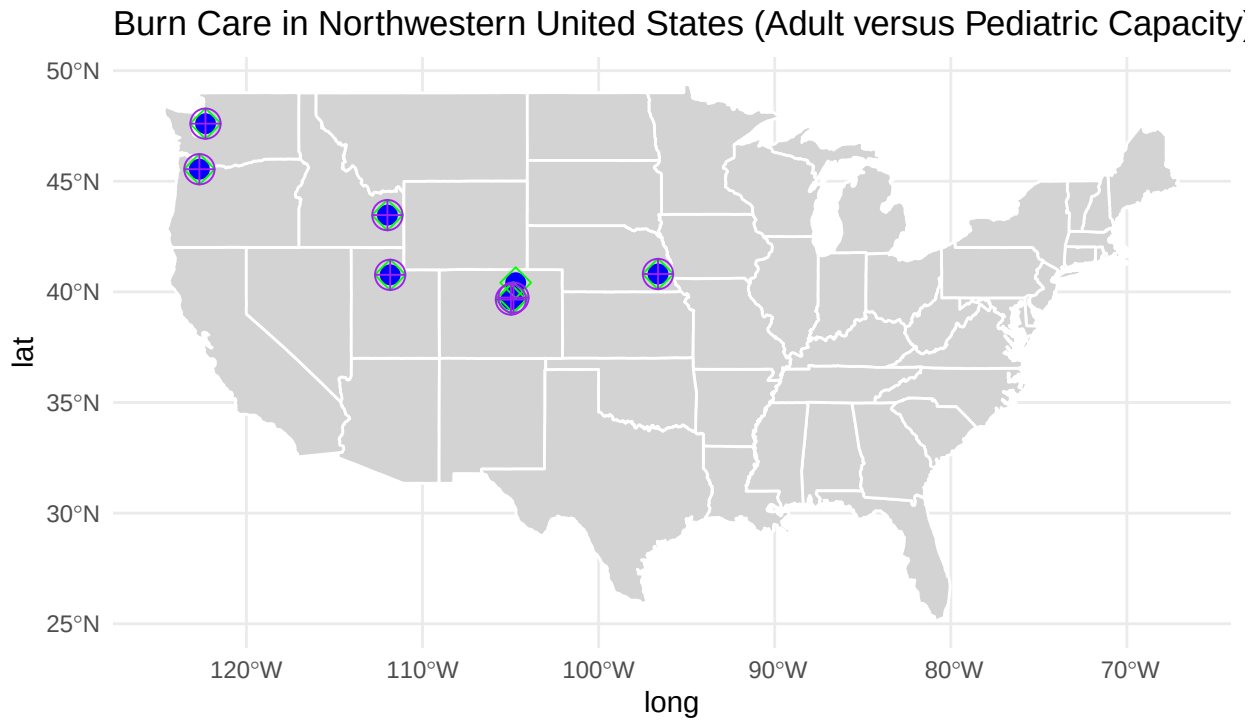
```
## Warning in min(cc[, 1], na.rm = TRUE): no non-missing arguments to min;
## returning Inf

## Warning in min(cc[, 2], na.rm = TRUE): no non-missing arguments to min;
## returning Inf

## Warning in max(cc[, 1], na.rm = TRUE): no non-missing arguments to max;
## returning -Inf

## Warning in max(cc[, 2], na.rm = TRUE): no non-missing arguments to max;
## returning -Inf
```

```r
##Plot Data
# Get usa map data
usa_map <- map_data("state")
#Plot Burn Capacity
ggplot() +
  # Plot the base map
  geom_polygon(data = usa_map, aes(x = long, y = lat, group = group), fill = "lightgray", color = "white
  # Add your location points
  geom_sf(data = loc_sf_beds, color = "blue", size = 2) +
  geom_sf(data = loc_sf_aba, color = "green", size = 4, shape = 23) +
  geom_sf(data = loc_sf_sd, color = "purple", size = 5, shape = 10) +
  # Set the coordinate limits and CRS for proper display
  coord_sf(crs = 4326) +
  labs(title = "Burn Centers in Northwestern U.S. Based on Designation Status") +
  theme_minimal()
```

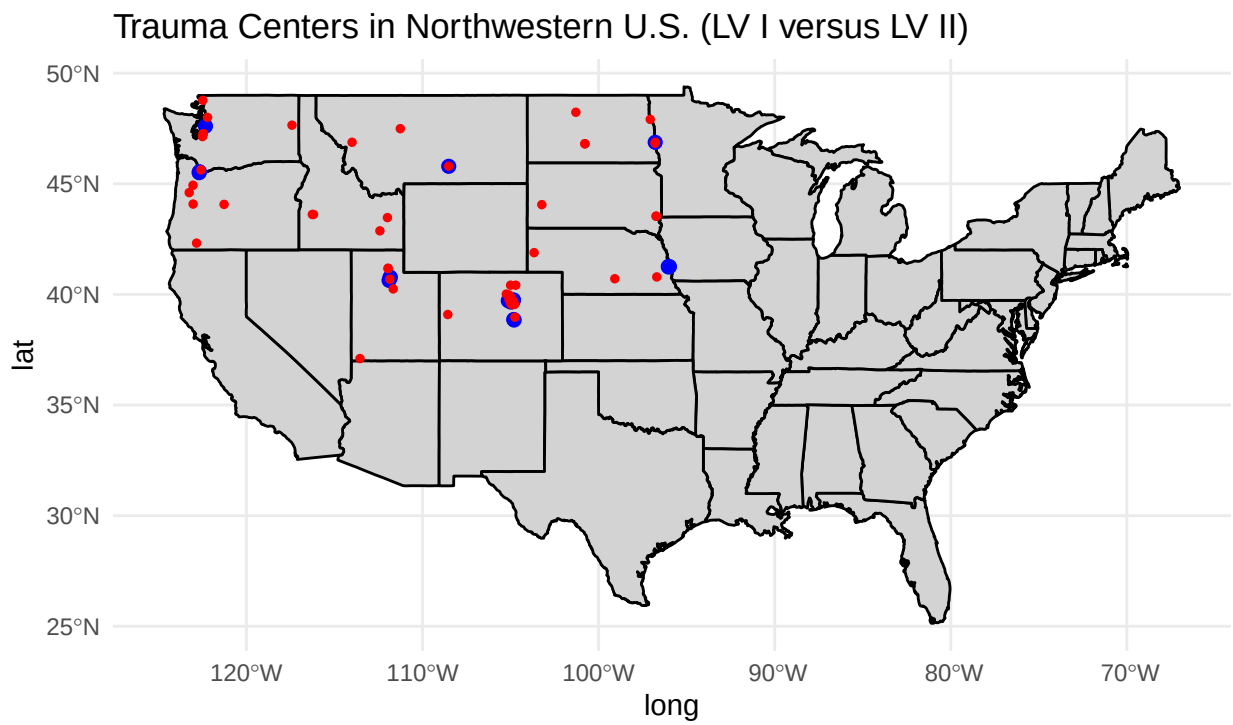



```r
##Plot Data
# Get usa map data
usa_map <- map_data("state")
#Plot Burn Capacity by Adults vs Peds
ggplot() +
  # Plot the base map
  geom_polygon(data = usa_map, aes(x = long, y = lat, group = group), fill = "lightgray", color = "white
  # Add your location points
  geom_sf(data = loc_sf_burn, color = "blue", size = 3) +
```

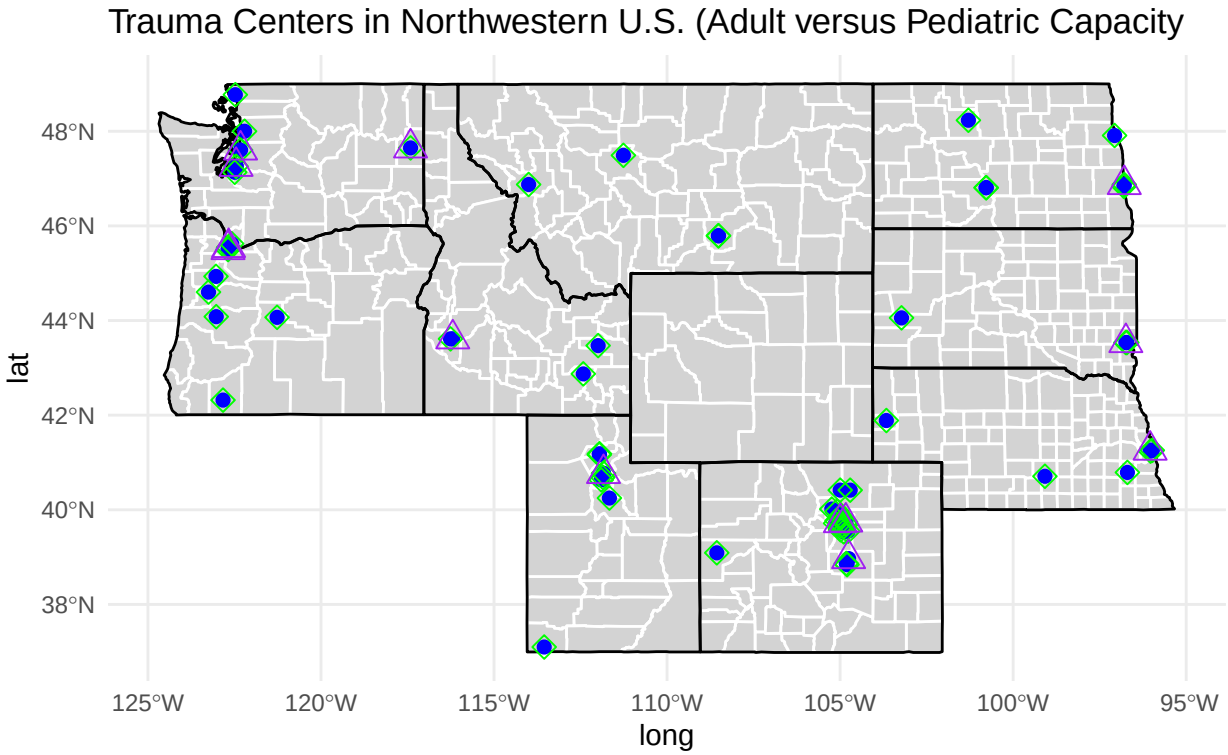
```
  geom_sf(data = loc_sf_burn_adults, color = "green", size = 4, shape = 23) +
  geom_sf(data = loc_sf_burn_peds, color = "purple", size = 5, shape = 10) +
  # Set the coordinate limits and CRS for proper display
  coord_sf(crs = 4326) +
  labs(title = "Burn Care in Northwestern United States (Adult versus Pediatric Capacity)") +
  theme_minimal()
```




```
##Plot Data
# Get usa map data
usa_map <- map_data("state")
#Plot Trauma Capacity
ggplot() +
  # Plot the base map
  geom_polygon(data = usa_map, aes(x = long, y = lat, group = group), fill = "lightgrey", color = "blac
  # Add your location points
  geom_sf(data = loc_sf_tc1, color = "blue", size = 2) +
  geom_sf(data = loc_sf_tc2, color = "red", size = 1) +
  # Set the coordinate limits and CRS for proper display
  coord_sf(crs = 4326) +
  labs(title = "Trauma Centers in Northwestern U.S. (LV I versus LV II)") +
  theme_minimal()
```




```
##Plot Data
# Get usa map data
regions <- "Montana"
regions <- c("Idaho", "Colorado", "Wyoming", "Utah", "Washington", "Oregon", "South Dakota", "North Dak
county_lines <- map_data("county", region = regions)
mi_counties <- subset(county_lines, region == "Montana")
target_counties <- c("Missoula", "Flathead", "Lake", "Ravalli")
mi_counties$highlight <- ifelse(mi_counties$subregion %in% target_counties, "Target", "Other")
state_lines <- map_data("state", region = regions)
#Plot Trauma Capacity
ggplot() +
  # Plot the base map
  geom_polygon(data = county_lines, aes(x = long, y = lat, group = group), fill = "lightgrey", color =
  geom_polygon(data = state_lines, aes(x = long, y = lat, group = group), fill = NA, color = "black") +
  # Add your location points
  geom_sf(data = loc_sf_tc12, color = "blue", size = 2) +
  geom_sf(data = loc_sf_tc_adults, color = "green", size = 3, shape = 23) +
  geom_sf(data = loc_sf_tc_peds, color = "purple", size = 4, shape = 24) +

  # Set the coordinate limits and CRS for proper display
  coord_sf(crs = 4326) +
  labs(title = "Trauma Centers in Northwestern U.S. (Adult versus Pediatric Capacity") +
  theme_minimal()
```


Trauma Centers in Northwestern U.S. (Adult versus Pediatric Capacity
48°N
46°N
44°N
42°N
40°N
38°N
lat
125°W
120°W
115°W
110°W
105°W
100°W
95°W
long